



NeuraScan AI

Intelligent Behavioural Screening for Neurological Risk Analysis

A four-minute smartphone session → nine on-device behavioural features → **deviation from the user's own baseline**, explained domain by domain. Offline-first, and never a diagnosis.

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ABSTRACT

Alzheimer's disease, Parkinson's disease and mild cognitive impairment alter memory, speech and movement years before diagnosis, yet screening stays clinic-bound and is judged against population norms that hide individual differences. **NeuraScan AI** turns a four-minute smartphone session into nine behavioural features, compares them with the user's own frozen baseline, and raises an explainable, non-diagnostic status only when the change persists. On a simulated cohort of 2,400 users it detected **98.8%** of gradual cognitive declines at **4.0%** false alerts.

PROBLEM STATEMENT

- **A long, unmonitored window.** Memory, speech fluency, fine motor control and reaction speed change years before a formal diagnosis.
- **Episodic.** A paper screen such as the MoCA is taken once, in a clinic, usually only after symptoms are noticed.
- **Population-referenced.** A high performer can decline far before crossing an age cut-off; a naturally slower person is flagged falsely.
- **Single-modality and hard to reach.** Most tools measure cognition *or* movement *or* speech, and specialists sit in cities.

OBJECTIVES

- 01 Deliver a standardised **4-minute screening session** on Android and iOS from one Flutter codebase.
- 02 Extract **9 interpretable features** on the device from touch, microphone and keystroke timing.
- 03 Build a **personal baseline** and a four-domain deviation engine with smoothing and a persistence rule.
- 04 Suppress bad-day false alarms with a **context check-in** and task-level quality gates.
- 05 Present an **explainable status** with per-domain contributions and a shareable PDF report.
- 06 Keep data private: encrypted offline-first storage, **opt-in sync of derived scores only**.

SYSTEM ARCHITECTURE

ALL SCORING HAPPENS ON THE DEVICE

PRESENTATION LAYER · FLUTTER UI



SENSING LAYER



ON-DEVICE SCREENING ENGINE · DART

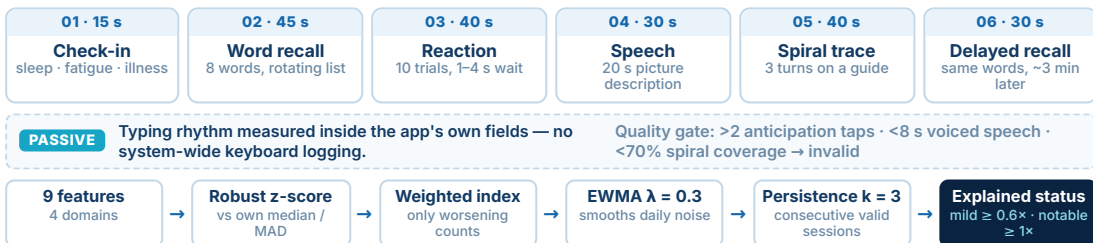


DATA LAYER



Raw audio, touch traces and typed text **never leave the phone** · owner-only Firestore rules · aligned with India's DPDP Act, 2023

METHODOLOGY — SESSION & DECISION WORKFLOW



Baseline: the first 2 sessions are discarded as practice; the next 6 valid sessions fix a median and MAD per feature, which are then frozen. Sessions the check-in marks as a poor night, illness or heavy fatigue are shown to the user but excluded from the baseline and the trend.

BEHAVIOURAL FEATURE MODEL

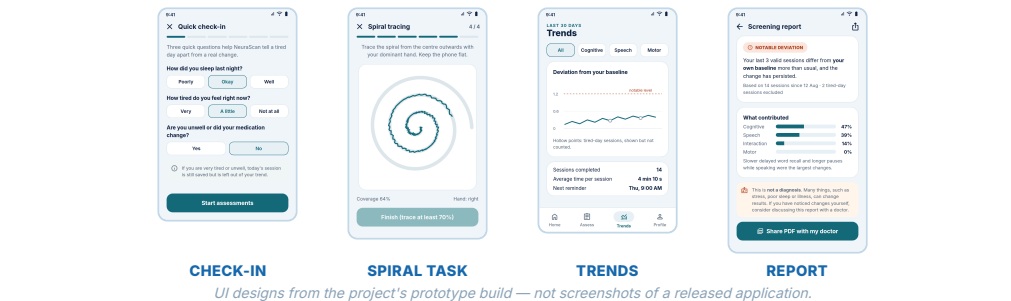
9 FEATURES · 4 DOMAINS

COGNITIVE 35%	SPEECH 25%	MOTOR 25%	INTERACTION 15%
Cognitive 35% Delayed recall ▼ words recalled after ~3 min ÷ 8 · Reaction median ▲ of 10 trials · Reaction CV ▲ SD ÷ mean	Speech 25% Speaking rate ▼ syllable nuclei / min · Pause ratio ▲ silence ≥ 250 ms ÷ total	Motor 25% Spiral RMSE ▲ radial error from the guide · Tremor index ▲ 4–12 Hz share of error power	Interaction 15% Inter-key interval ▲ median gap between inserts · Inter-key CV ▲ SD ÷ mean

▼ worse when lower · ▲ worse when higher · each feature becomes a robust z-score against the user's own baseline.

IMPLEMENTATION & TECHNOLOGY STACK

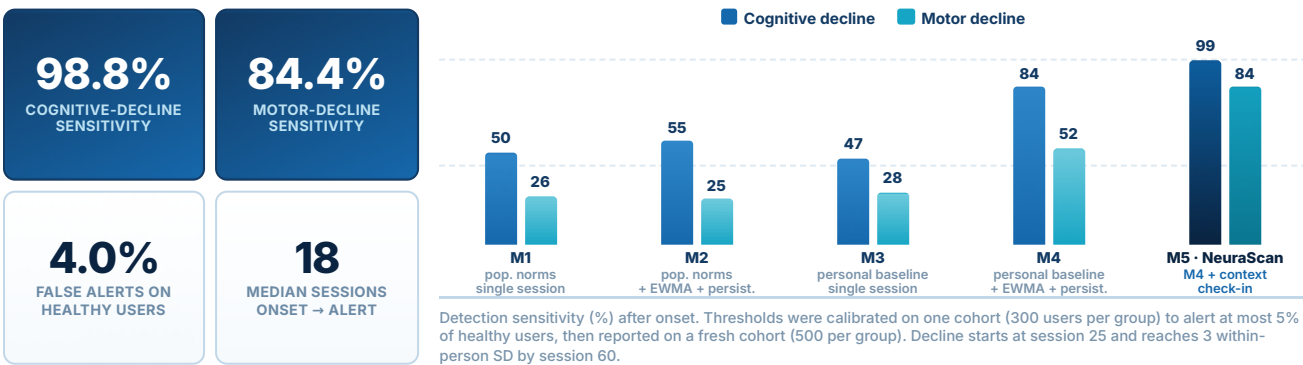
- **Ten Flutter screens** — Material 3, Riverpod state, go_router navigation, English and Tamil interfaces.
- **Pure-Dart engine module**, mirrored by a Python reference implementation; **12 / 12 unit tests pass**.
- **Five Drift tables** behind SQLCipher; the same entities nest under `/users/{uid}` in Firestore, so one owner-only rule covers them all.
- **Explanation is exact, not approximate** — each domain's share of the index *is* its contribution; the four shares sum to 100%.



CHECK-IN SPIRAL TASK TRENDS REPORT
UI designs from the project's prototype build — not screenshots of a released application.

RESULTS — SIMULATED-COHORT EVALUATION

2,400 SYNTHETIC USERS · 60 SESSIONS EACH · MATCHED FALSE-ALERT RATES



METHOD	FALSE ALERTS	SENS. COG.	SENS. MOTOR	DELAY COG/MOT
M1 Population norms, single session	5.6%	50.0%	26.4%	24 / 24
M2 Population norms + EWMA + persistence	5.6%	55.4%	24.8%	25 / 27
M3 Personal baseline, single session	3.4%	46.6%	28.4%	23 / 25
M4 Personal baseline + EWMA + persistence	4.4%	84.2%	52.0%	23 / 25
M5 NeuraScan (M4 + context check-in)	4.0%	98.8%	84.4%	18 / 23
5.4 μs engine update 50 ms target	20 numbers of baseline state	≈0.5 KB synced / session no raw signals	12/12 engine unit tests passing	

KEY FINDINGS

- **Population norms miss most declines.** A user who starts well above average can lose a great deal and still look normal — M1 catches only 50.0% / 26.4%.
- **A personal baseline alone is not enough.** Judged session by session (M3), noise forces a high threshold and sensitivity stays at 46.6%.
- **How change is judged matters as much as what is measured.** The same nine features rise from 50.0% to 98.8% once the baseline is personal and the change must persist.
- **Persistence helps the noisier motor domain most.** k = 5 reaches 99.4% / 91.6%, but adds about ten days of waiting, so k = 3 is the default.
- **Decline is rarely single-domain.** The top contributor matched the simulated domain in 66.4% / 68.2% of alerts, so all four shares are shown rather than one label.
- **The honest limit is the evidence.** Every figure above comes from synthetic users with parameters we chose. The method is cheap enough for a mid-range phone (5.4 μs per update), but clinical validity still needs a real pilot.

CONCLUSION

NeuraScan AI shows that a mobile application can make neurological risk screening frequent, personal, explainable and private. The central result is that **how a phone judges change matters as much as what it measures**: the same nine features detected 50.0% of simulated cognitive declines against population norms and 98.8% against a personal baseline with persistence and context gating, at matched false-alert rates.

Scope: a screening and awareness tool, not a diagnostic device. All quantitative results come from a simulated cohort — clinical validity is unproven.

FUTURE ENHANCEMENTS

- **Pilot study** with volunteers under institutional ethics approval — test–retest reliability and usability.
- **Guarded re-baselining** after long stability, plus per-device baselines.
- **On-device TensorFlow Lite** speech features, anomaly detection and learned weights, with SHAP keeping explanations.
- **Gait and postural tremor** tasks using the accelerometer and gyroscope.
- **Clinician web view** receiving a report only with the user's consent.
- **Smartwatch sensing and FHIR export**; federated learning so models improve without collecting raw data.

SELECTED REFERENCES

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Full list of 20 references appears in Chapter 13 of the project report.